Small Head Sets for Simple Tasks

Lexical task heads, function-vector heads and attention filters in Llama-3.1-8B-Instruct

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Abstract

We look for the smallest set of attention heads whose mean output, written into a prompt that does not state the task, makes Llama-3.1-8B-Instruct perform it, on 14 simple tasks. We start from two published ways of choosing heads: lexical task heads, found by decoding each head's output into words (Yang et al., 2026), and the heads of instruction function vectors, found by their causal effect (Davidson et al., 2025). We re-implement both, read heads with the per-head logit lens and with Head-JLens, and then compare a series of selection rules under one held-out protocol. Head-JLens finds more heads that carry the task than the logit lens. The hand-written term list is the weak point of the lexical method. Lists built from the words heads actually decode, together with filters on where a head attends, come close to causally chosen sets without patching anything during selection. Different task types rest on different heads: causal rankings carry morphology and sequences, attention to the instruction's task words carries translations, and facts are carried by almost any good set, sometimes by one head. The best set we found is shared by all tasks: Davidson et al.'s function vector without the few heads that add the same direction on every task. With about two dozen heads it carries most tasks better than per-task sets several times its size.

1 Introduction

A task representation in a language model can be located by finding the attention heads whose outputs, averaged over prompts that state the task and written into a prompt that does not, make the model perform the task (Todd et al., 2024). Two recent studies choose those heads in different ways. Yang et al. (2026) decode each head's output at the last prompt token with the logit lens and call a head a *lexical task head* when its top tokens contain words describing the task (*opposite*, *capital*). Davidson et al. (2025) rank heads by their causal effect on the answer when the prompt's instruction is replaced by uninformative text, and sum the mean outputs of the top heads into an *instruction function vector*.

Our goal is the smallest set of heads that represents the tasks well, ideally one set for all of them. We reproduce both methods on the same model and tasks, add a second readout, Head-JLens (Merlo, 2026), and then test a sequence of selection rules built on what we learnt: word lists taken from what heads decode, the source methods' head rankings at fixed sizes, filters on where a head attends, and sets shared across tasks. All rules are compared under one protocol, with heads chosen on one half of each task's items and tested on the other.

The paper is organised linearly: the common setup (Sec. 2), the two reproductions (Sec. 3), the selection rules (Sec. 4), their results (Sec. 5), and the approaches we would keep (Sec. 6). Only simple tasks are covered.

2 Setup

Model and tasks. Llama-3.1-8B-Instruct in bfloat16, the main model of Yang et al. (2026) and the only one with released Head-JLens matrices. The 14 simple tasks shared by both source studies, with the datasets of Todd et al. (2024): antonym, synonym, singular–plural, present–past, country–capital, country–currency, landmark–country, park–country, product–company, next-item, prev-item, English–French, English–German, English–Spanish. We group them as word relations, morphology, facts, sequences and translations. Prompts follow Yang et al. (2026): raw text without the chat template; five instruction templates per task (the cloze form "The antonym of {x} is" and four of the form "<instruction> Q: {x}\nA:"); example prompts with 1 to 30 shots (" x_1: y_1; x:"); and the zero-shot prompt " x:", which states no task. Decoding is greedy.

Reading a head. For head h of layer l, $z_{l,h}$ is its output at the last prompt token before the output projection and $c_{l,h} = W_O^{(h)} z_{l,h}$ its write to the residual stream. With q_v the unembedding row of token v scaled by the final norm's gain, the two readouts score v by

$$\text{Logit-head: } q_v^\top c_{l,h}, \qquad \text{J-head: } q_v^\top J_{l,h}\, z_{l,h},$$

where $J_{l,h}$ is the head's part of the Jacobian map of Head-JLens, fitted on WikiText-103 (Merlo, 2026). J-head reads the head through the layers that follow it; Logit-head is the readout of Yang et al. (2026). A head *decodes* a word on a prompt when one of the word's token forms is among its 10 highest-scoring tokens.

Testing a head set. A set is tested by mean patching (Yang et al., 2026, Sec. 2.3.2). For each head in the set, its mean output over prompts that state the task (correctly answered, items not patched) is computed; at the last token of the patched prompt, the head's output is replaced by α times this mean, and the model generates its answer. There are two checks.

- **Check 1** patches the zero-shot prompts of items the model answers correctly with one shot. The prompt says nothing about the task, so unpatched accuracy is near zero, and a gain means the set brings the task in. This is the main test.
- **Check 2** patches the 1-shot prompts of items the model answers *wrongly* with one shot (unpatched accuracy 0). The prompt already shows the task and the answer format, so this check measures repair rather than transfer.

In both checks a *control* patches the same heads with the mean of a related task (the pairing of Yang et al., 2026, e.g. antonym with synonym); a set carries its task when its own mean helps and the control mean does not.

Scoring. A reply is correct when it begins with the whole reference answer as words, ignoring case, accents, punctuation and a leading article. This replaces the first-token match of Yang et al. (2026), which marks any number correct on next- and prev-item and a capitalised German noun wrong. The same score selects the correctly answered prompts used everywhere else.

Held-out protocol. From the comparison of selection rules on (Sec. 5), every quantity that depends on items is chosen on one half of each task's items and tested on the other. The items are shuffled once (seed 0) and split into halves A and B; fold 1 chooses on A and tests on B, fold 2 the reverse. "Chosen" covers the prompts whose heads are read, word lists, head sets, rankings, the causal scores of Davidson et al., and the task and control means. The means are rule-style: one per instruction template, over its first 100 correctly answered prompts of the choosing half; each patched item is patched with each of the five means and accuracy is averaged over them. Up to 100 items per task and fold are patched in check 1, and up to 100 in check 2 (as few as 11 on some tasks). The scaling is $\alpha \in \{1, 2\}$. For each fold, α is the value that was better on the other fold for the same task and set (cross-fitting; ties go to $\alpha = 1$), so it is never chosen on the items it is scored on. Random sets are scored at the α of the set they are matched to. Reported numbers are means of the two folds. Replies stop at a newline, the example delimiter or 16 tokens. Every run is single and greedy, and we report accuracies only, without intervals.

3 Reproducing the two source methods

3.1 Lexical task heads (Yang et al., 2026)

Method. A head is a lexical task head for a task and prompt style if, on at least 10% of 100 correctly answered prompts, at least one of its top-10 Logit-head tokens is in a hand-written term list for the task (2 to 22 tokens). The prompts are those of the most accurate instruction template (rule style) or the 5-shot prompts (example style). The set is then tested by mean patching with each wording's own mean, at 15 values of α from -10 to 10.

Fidelity. We re-run their code's data, templates, term lists, control pairs and parameters (Table 1). We depart from it in four places, each by choice: the padding mask, the J-head readout, the answer score, and bfloat16 instead of float32, the precision in which the lens was fitted. Their code passes batches of 20 left-padded prompts without an attention mask, so the model attends to the padding; we mask it. Run in their code's own setting (padding attended, first-token score), our Logit-head counts are within one head of their Figs. 79 and 88 on every task (rule style: 250 heads over the 14 tasks against their 247; example style: 127 against 130).

Setting	Yang et al. (2026)	Here
Model, precision	Llama-3.1-8B-Instruct, float32	same model, bfloat16
Tasks, templates, term lists	14 Todd et al. tasks; first 5 of the 6 templates; their lists	same
Batching	20 left-padded prompts, padding attended	same batches, padding masked
Readout	Logit-head	Logit-head and J-head
Head criterion	top 10, $\geq$1 term, $\geq$10% of the first 100 correct prompts	same
Answer score	first generated token	generated reply, flexible score
Means, α, control	one mean per wording; 15 values $-10..10$; their control map	same
Check 1	zero-shot prompts (paper, App. G.1); no released script	as the paper
Check 2	1-shot prompts answered wrongly (the released script)	same; control added in the held-out runs

Table 1: Lexical task heads: their setting and ours.

What reproduces. Logit-head finds lexical task heads for every task. In check 1, which their paper describes, patching gains only on the fact tasks (and a few more with example means) and does nothing on antonym, synonym, present–past, prev-item and the translations. The per-task curves of their App. G.1 (Figs. 57 and 62), which the text attributes to zero-shot patching, reproduce only in check 2, the setting of their one released causal script. There every task gains, with the largest and smallest gains on the same tasks as in their figures. Their headline finding thus holds in check 2. In check 1, the lexical heads carry only some task types.

J-head against Logit-head (Table 2). J-head finds 50 to 70% more heads; most of the added ones lie in layers 10–19, where Logit-head finds few. The heads only J-head finds have causal effects of their own, and J-head sets are at least as effective as Logit-head sets in both checks and both styles. The lead is not a matter of set size: comparing the top 5, 10 and 20 heads of each readout, most of J-head's leads remain and none reverses, while none of Logit-head's few leads remains. J-head does not contain the Logit-head set (only on 4 of 28 task–style pairs). From here on all sets are read with J-head. The one later comparison of readouts, LLM-scored heads on two tasks, mostly points the same way.

	Heads (sum over tasks)			Check 1 effect			Check 2 effect		
Style	L	J	L∩J	L	J	J only	L	J	J only
Rule	240	370	199	0.17	0.19	0.10	0.26	0.32	0.27
Example	141	237	107	0.22	0.32	0.24	0.33	0.43	0.38

Table 2: Lexical task heads under both readouts (items not held out). Heads: summed over the 14 tasks. Effect: highest accuracy over the 15 values of α, averaged over the style's five means, minus unpatched accuracy; mean over tasks. J only: heads J-head finds and Logit-head does not.

3.2 Instruction function vectors (Davidson et al., 2025)

Method. Each head is scored by its causal indirect effect: the gain in probability of the answer's first token when that head alone is set to its mean output, on prompts whose instruction is replaced by uninformative text of matched length and log-probability. There are three kinds of such text: token sequences sampled to be as probable as the instruction, WikiText-103 prefixes, and other tasks' instructions. The score is averaged over kinds and tasks, and the top 20 heads form one set for all tasks. A task's function vector (FV) is the sum of those heads' mean outputs, added to the residual stream after layer 11 ($|L|/3$) with weight 1.

Fidelity. We implement the method from the paper; no public code exists, and the instruction-FV accuracies for this model appear only in a figure, so we have no published per-task numbers to compare with. Table 3 lists the differences. The main ones are forced by our setting: our instructions are Yang et al.'s templates, and the FV is evaluated in our check 1.

Setting	Davidson et al. (2025)	Here
Model, tasks	several models incl. Llama-3.1-8B-Instruct; 50 Todd et al. datasets	Llama-3.1-8B-Instruct; our 14 tasks
Instructions	5 best of LLM-generated instructions, short and long	Yang et al.'s templates 1–4 (template 0 has no separate instruction)
Mean outputs	100 correct prompts, 20 per instruction	100 correct prompts, 25 per template
Uninformative text	3 kinds, matched in length and log-probability; 25 prompts per kind; 2 instruction lengths	same 3 kinds and matching; 20 prompts per kind; one length; smaller WikiText pool
Head score	first-token probability gain, averaged over kinds and tasks	same
Heads in the FV	top 20, one set for all tasks	top 20 (4-task run); top 30 over the 14 tasks (later runs)
Intervention	sum of means added after layer 11, weight 1	same at the last prompt token; also other weights
Evaluation	zero-shot, no instruction, their prompt	check 1 (" x:"), flexible score, control mean

Table 3: Instruction function vectors: their setting and ours.

What we find. The FV works at their weight: accuracy peaks at $\alpha = 1$ and vanishes from $\alpha = 3$ on. With the 30 heads chosen over the 14 tasks, it beats our best lexical-head set on 11 of 14 tasks, including tasks the lexical sets do not carry (singular–plural, antonym). It does nothing on present–past, prev-item and synonym and little on the translations. On held-out items it is about level with the decoded-word sets and below the best sets found later (Table 5). Reading its 30 heads with J-head shows why the weight is so constrained. Six of them decode the same tokens on every task (line ends, colons), attend to the answer slot, and add one large direction shared by all tasks. Without these six (24 heads) the FV is at least as good on all 14 tasks, works best at α 2–3, and also carries the translations. Removing six random FV heads instead does not help. On held-out items, the FV without its generic heads is the best set we found (Sec. 5.3).

Two further sets come from Davidson et al.'s score: **FV-N**, the top N heads by the score on each task alone, and **FV-univ-N**, the top N by the score averaged over the 14 tasks (their choice of heads). Both are patched head by head, like every other set here, rather than summed into the residual stream.

4 Selection rules

Table 4 lists the rules compared in Sec. 5, all on J-head readings of the rule-style prompts of the choosing half (five templates pooled, about 500 prompts per task). The rules differ in what they require of a head and in what they need: the lens only, attention patterns (one forward pass per prompt), or a causal score (one patched forward pass per head and prompt, 1,024 heads).

Decoded-word lists. A word is any decoded top-10 token that, stripped and lower-cased, is alphabetic, has at least 3 letters, is not a stop word and is common English (Zipf frequency $\geq$3). Its *share* on a task is the fraction of (prompt, head) pairs whose top 10 holds it. *Task words* have the largest share on the task minus their mean share on the other 13; *common words* have the largest minimum share over all 14 (*what*, *which*, *maybe*, *https*). *Most active words* have the largest share, with no contrast. A head passes a list when it decodes one of its words on at least 10% of the pooled prompts, the threshold of Yang et al. (2026).

Attention filters. At the last prompt token, a head's mean attention over the task's prompts on four parts of the prompt: the BOS token (B: pass if ≤ 0.5); the instruction's task words, i.e. Yang et al.'s terms and the task's 10 top J-head task words (T: pass if ≥ 0.05); the query (I: ≥ 0.05); and the rest of the instruction (R: ≤ 0.5). D is decoding: the head decodes one of the task's 30 most active words on at least 10% of prompts. All words are matched with and without a leading space, in lower, capitalised and upper case, because the lens decodes names as " Japan". A set is named by the filters a head must pass; all 31 combinations were run.

The order in which the rules were tried explains their shape. E1 (Yang et al.'s rule under J-head) carried few task types. Replacing the hand-written list by decoded words (CC-pooled) doubled the effect but needed about 120 heads, and its top 20 or 40 by rate did little. Davidson et al.'s ranking did better at every size, and reading its heads showed that the decoding heads which work attend to the prompt, whereas most others look at BOS (E7). E8 crossed decoding with four attention filters. E10 combined E8's scores over tasks into

Set	A head is kept if ...	Needs
One set per task		
E1	it decodes a term of Yang et al.'s hand-written list on $\geq$10% of the best template's prompts	lens, list
CC-pooled	it decodes one of 5 task + 5 common words on $\geq$10% of the pooled prompts	lens
CC-N	top N by that rate	lens
E7 words + B	it decodes one of the 10 most active words ($\geq$10%) and attends to BOS $\leq$0.5	lens, attention
E6 instruction diff.	CC rule applied to clean minus corrupted-instruction outputs	lens
FV-N	top N by Davidson et al.'s score on the task	causal
E8 D, DB, DBT, BT, T	it passes every named filter (D decoding, B BOS, T task-word attention)	lens and/or attention
One set for all 14 tasks		
Davidson FV-30	top 30 by the score averaged over tasks; their intervention (summed means added after layer 11) at $\alpha = 1$	causal
no generic heads	the same 30 minus the generic heads among them (5; 25 heads left); their intervention	causal
FV-univ-N	top N by the score averaged over tasks, patched head by head; FV-univ-30 = the heads of FV-30	causal
E10 shared DBT-N	per filter, each head's within-task rank averaged over tasks; D, B, T ranks averaged; top N (BT: mean scores)	lens, attention
Controls		
Random-N	as many heads per layer as CC-N, drawn outside it (5 draws)	–
Random from E7	as many heads as E7 words + B, drawn from E7's unfiltered set (5 draws)	–
Random, DB/DBT/T layers	as many heads per layer as the E8 set, drawn outside it where the layer allows (5 draws)	–
Random from D	as many heads as DB or DBT, drawn from the task's D set (5 draws)	–
Random shared	as many heads per layer as FV-univ-20 or E10 shared DBT, drawn outside it, one draw for all tasks (5 draws)	–

Table 4: Head-selection rules compared under the held-out protocol. E-numbers name the experiment in which each was introduced.

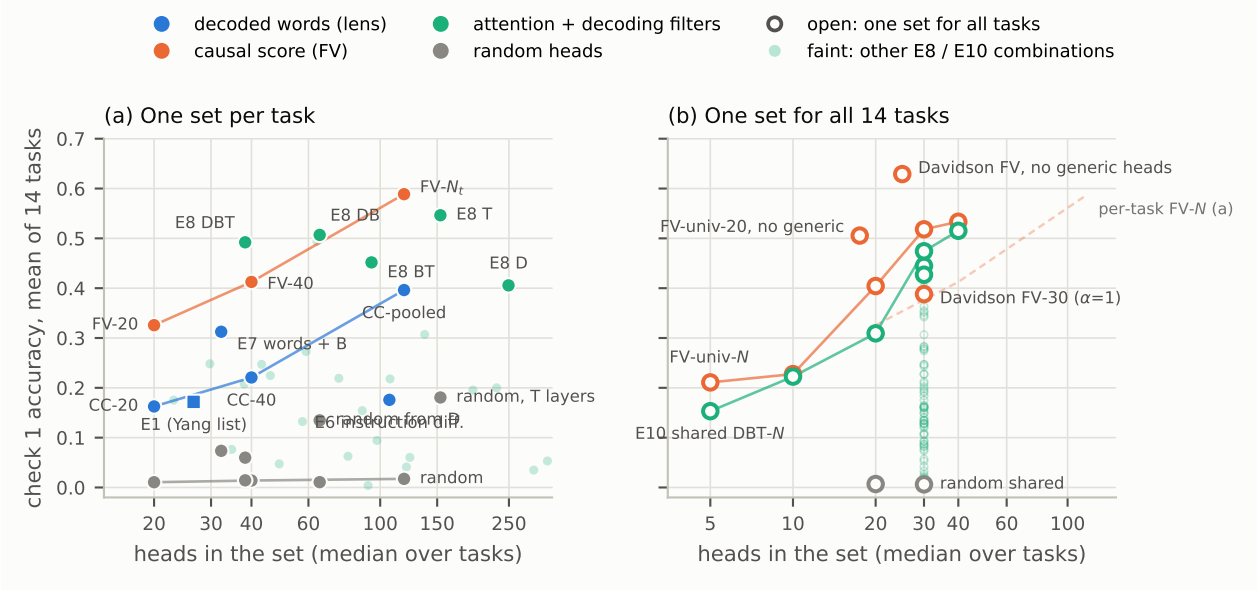


Figure 1: Check 1, mean accuracy over the 14 tasks against the number of heads (median over tasks; log scale), held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. (a) One set per task; (b) one set for all tasks. Colour: how heads are scored; filled: per task; open: shared; square: E1, Yang et al.'s term list. Lines join the sets of one ranking at increasing size. Grey: random controls (unlabelled points: random from E7, and layer-matched to DB and DBT). Faint dots: the other E8 (a) and E10 (b) filter combinations. Dashed in (b): the per-task FV-N line of (a), for reference.

one set of 30 heads, the size of Davidson et al.'s set, for a direct comparison with theirs. E11 completed the comparison: the shared rankings at 5 to 40 heads, random controls for every kind of set, Davidson et al.'s intervention in both checks with and without its generic heads, and both folds for every set.

5 Results

5.1 Accuracy against set size

Figure 1 and Table 5 give every set's mean over the 14 tasks.

- **Which heads matters, not how many.** Random heads with the same number of heads per layer do nothing, at any size, per task or shared. Random decoders do little: as many heads drawn from D as DB or DBT holds reach a fraction of the filtered sets, and the same holds for E7. The one random control above zero is the one matched to T, whose sets fill whole layers so that the draws share many of T's heads. Control means stay low for every set in check 1, so the gains are task-specific.
- **Hand-written lists are the bottleneck of the lexical method.** Decoded-word lists (CC-pooled) more than double E1's mean, but only at about 120 heads; their top 20 or 40 by rate are no better than E1. Heads chosen on what the instruction adds (E6) are worse than CC at full size: the heads they drop decode the task's words with and without the instruction, and check 1 needs them.
- **A causal ranking is better at every fixed size** (FV-20, FV-40, FV-N_t against CC-20, CC-40, CC-pooled), and FV-N_t has the highest mean of the per-task sets. It grows steadily with size.
- **Attention filters close most of the gap without patching.** Keeping the decoding heads that do not look at BOS (DB) or that also attend to the task words (DBT) gives sets of about 40 to 65 heads above FV-40 and below FV-N_t. Attention to the task words alone (T) comes close to FV-N_t, but with more heads. Attention to the query or to the rest of the instruction lowers every combination it is added to (faint dots).
- **A shared set does best overall** (Sec. 5.3): Davidson et al.'s FV without its generic heads, with about a fifth of FV-N_t's heads per task.

5.2 By task type

The means hide large differences between task types (Figure 2, Table 6; per task in App. A).

- **Facts** (capitals, currency, landmark, park, product) are carried by almost every good set, shared or not, best by the FV without generic heads. E7 is the exception on landmark–country: its patched replies begin

Head set	Heads	Check 1		Check 2	
	median (range)	task	control	task	control
One set per task					
Decoded words (lens)					
E1: Yang term list	26 (8–48)	0.17	0.03	0.27	0.09
CC-20	20	0.16	0.03	0.27	0.10
CC-40	40	0.22	0.05	0.31	0.10
CC-pooled	118 (98–146)	0.40	0.05	0.34	0.13
E7 raw words	110 (80–134)	0.34	0.04	0.32	0.10
E7 raw words + B	32 (18–42)	0.31	0.02	0.34	0.13
E6 instruction diff.	107 (84–128)	0.18	0.01	0.33	0.10
Causal score (Davidson et al.)					
FV-20	20	0.33	0.03	0.34	0.10
FV-40	40	0.41	0.03	0.39	0.12
FV-N_t	118 (98–146)	0.59	0.04	0.39	0.12
Attention and decoding filters					
E8 D	249 (210–274)	0.41	0.02	0.26	0.05
E8 DB	65 (52–84)	0.51	0.01	0.32	0.09
E8 DBT	38 (3–47)	0.49	0.03	0.35	0.14
E8 BT	94 (12–114)	0.45	0.02	0.38	0.13
E8 T	154 (13–302)	0.55	0.02	0.40	0.11
Random heads					
Random-20	20	0.01	–	0.08	–
Random-40	40	0.01	–	0.10	–
Random-N_t	118 (98–146)	0.02	–	0.12	–
Random from E7 raw	32 (18–42)	0.07	–	–	–
Random from D, DB size	65 (52–84)	0.13	–	0.18	–
Random, DB layers	65 (52–84)	0.01	–	0.08	–
Random from D, DBT size	38 (3–47)	0.06	–	0.14	–
Random, DBT layers	38 (3–47)	0.01	–	0.08	–
Random, T layers	154 (13–302)	0.18	–	0.16	–
One set for all 14 tasks					
Causal score (Davidson et al.)					
Davidson FV-30 (α=1)	30	0.39	0.05	0.34	0.16
Davidson FV-30, no generic heads	25	0.63	0.05	0.41	0.13
FV-univ-20	20	0.40	0.03	0.34	0.12
FV-univ-40	40	0.53	0.04	0.38	0.14
FV-univ-20, no generic heads	18	0.51	0.04	0.39	0.14
Attention and decoding filters					
E10 shared DBT	30	0.47	0.01	0.38	0.10
E10 shared BT	30	0.44	0.02	0.38	0.15
E10 shared DB	30	0.43	0.02	0.36	0.18
Random heads					
Random shared, FV-univ-20 layers	20	0.01	–	0.06	–
Random shared, E10 DBT layers	30	0.01	–	0.06	–

Table 5: Mean accuracy over the 14 tasks, held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. Heads: median (range) over tasks. Control: the same heads with the control task's mean. Random sets: task mean only, at the α of the set they are matched to. Davidson FV-30: their intervention at their $\alpha = 1$; without its generic heads, cross-fitted like the rest. Unpatched accuracy: 0.01 (check 1), 0 (check 2). Sets run on a ten-value grid score higher at their best α there (CC-pooled 0.44, E7 words + B 0.40, FV-20 0.38, FV-40 0.43); the order of the leading sets does not change. The shared rankings at 5 to 40 heads are in Table 7.

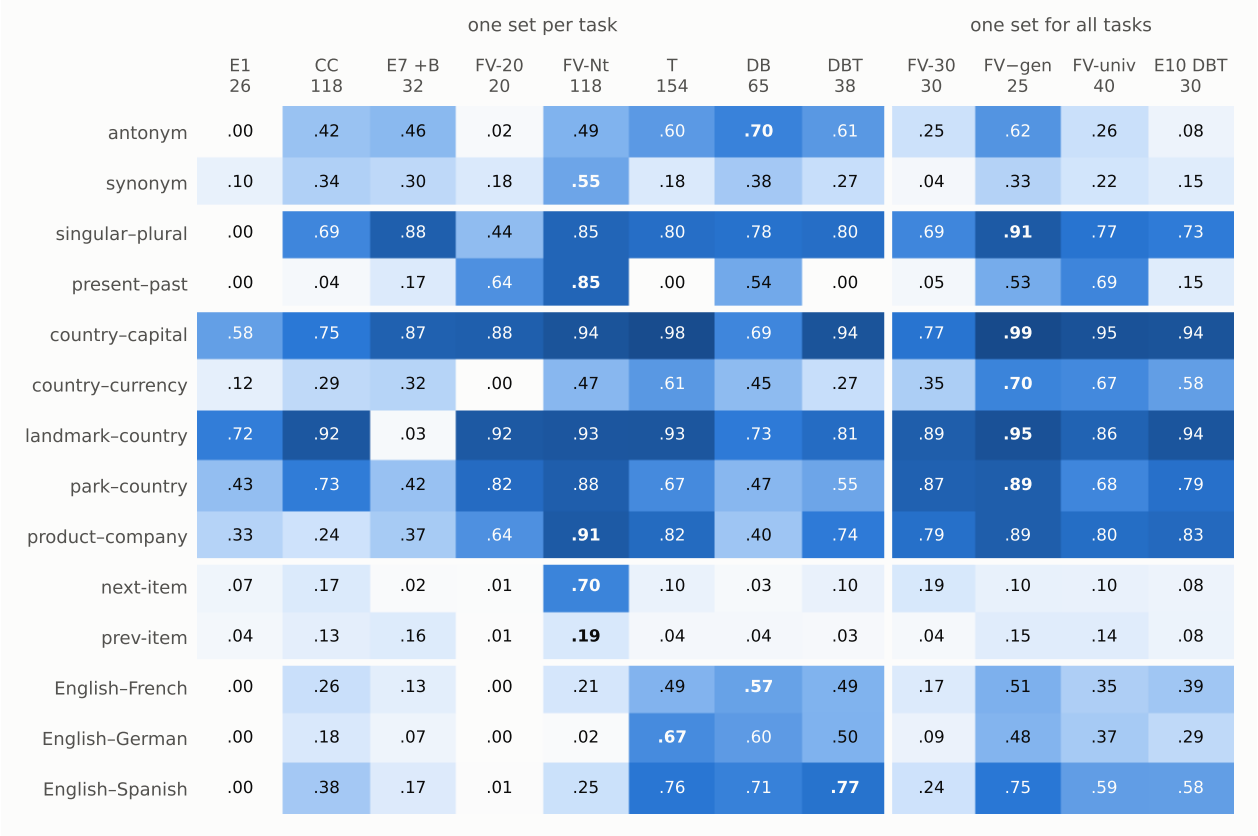


Figure 2: Check 1 per task, held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. Rows grouped by task type; column labels give the median number of heads. Bold: the row's highest value. Shared sets: Davidson FV-30 ($\alpha = 1$), the same without its generic heads (FV−gen), FV-univ-40, E10 shared DBT (30).

Task group	E1	CC	E7 +B	FV-20	FV-N_t	T	DB	DBT	FV-30	FV−gen	FV-univ	E10 DBT
Word relations (2)	0.05	0.38	0.38	0.10	0.52	0.39	**0.54**	0.44	0.14	0.48	0.24	0.12
Morphology (2)	0.00	0.36	0.53	0.54	**0.85**	0.40	0.66	0.40	0.37	0.72	0.73	0.44
Facts (5)	0.44	0.59	0.40	0.65	0.83	0.80	0.55	0.66	0.73	**0.89**	0.79	0.82
Sequences (2)	0.06	0.15	0.09	0.01	**0.44**	0.07	0.04	0.07	0.12	0.12	0.12	0.08
Translations (3)	0.00	0.28	0.12	0.00	0.16	**0.64**	0.62	0.59	0.17	0.58	0.44	0.42

Table 6: Check 1, mean accuracy per task type (number of tasks in brackets), the sets of Figure 2. Bold: highest in the row.

"Country:".

- **Morphology**: singular–plural is easy for most sets; present–past is carried by causally chosen heads (FV-20, FV-N_t, FV-univ, FV−gen) and, less well, by DB. T and DBT have almost no heads on present–past (13 and 3) and do nothing.
- **Sequences** (next-, prev-item) are carried only by FV-N_t, and prev-item only weakly. No decoding or attention filter and no shared set finds those heads.
- **Translations** are carried by the attention filters (T, DB, DBT) and by the FV without generic heads, and by no causally chosen per-task set. Removing the generic heads is what brings Davidson et al.'s FV to the translations.
- **Word relations** need many causally chosen heads (FV-N_t), the decoding filters (DB, T) or the FV without generic heads; FV-20 and the other shared sets carry synonym poorly.

No single per-task rule wins everywhere: FV-N_t leads on morphology and sequences, the attention filters on translations and word relations. The shared FV without generic heads is never far below the best set of a type, except on sequences.

5.3 One set for all tasks

The aim is a small set that represents the tasks in general. Figure 1b, Table 5 (lower part) and Table 7 give the shared sets.

	FV-univ-N			E10 shared DBT-N		
N	check 1	control	check 2	check 1	control	check 2
5	0.21	0.02	0.25	0.15	0.03	0.25
10	0.23	0.02	0.28	0.22	0.03	0.31
20	0.40	0.03	0.34	0.31	0.01	0.33
30	0.52	0.05	0.38	0.47	0.01	0.38
40	0.53	0.04	0.38	0.52	0.02	0.39

Table 7: The two shared rankings at 5 to 40 heads: check-1 task and control mean, check-2 task mean, over the 14 tasks; held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. FV-univ-30 holds the same heads as Davidson FV-30; E10 shared DBT-30 is E10's set.

- **Davidson et al.'s FV without its generic heads is the best set we found.** The FV-30 heads chosen on the choosing half hold five of the six generic heads in both folds. Removing them leaves 25 heads, one set for all tasks, which with Davidson et al.'s intervention at $\alpha = 2$ (the value chosen on the other fold for every task and fold) is above every per-task set in check 1, including FV-N_t and T with about five to six times as many heads per task. In check 2 it is level with the best sets, and its control is low in check 1.
- **The same 30 heads, used three ways.** Summed into the residual stream at Davidson et al.'s weight, the 30 heads are about level with the decoded-word sets, and at $\alpha = 2$ they collapse. Patched head by head (FV-univ-30) they are well above that and tolerate $\alpha = 2$. Summed without the generic heads they do best. So most of the gain over the published FV comes from the generic heads, whose shared direction limits how strongly the vector can be added. The remaining run, the 25 heads patched head by head, would tell how much the intervention itself adds.
- **Twenty heads are not enough to be safe.** The shared rankings rise steeply up to 30 heads; from 30 to 40, FV-univ changes little and E10 shared DBT still gains. At 5 or 10 heads they carry little. FV-univ-20 was the best set per head in fold 1 but much weaker in fold 2, where its top 20 holds a third generic head (L31H14). Without its generic heads it is better in both folds and steadier across them. At 30 and 40 heads the two folds agree.
- **Without patching, E10 shared DBT** is below FV-univ at every size and close to it only at 40 heads. It is chosen from J-head readings and attention alone, so selecting it needs no patching.
- **Random shared sets** with the same layers do nothing.

The two kinds of shared set converge: E10 shared DBT (30) holds 12 of fold 1's FV-univ-20 heads, all in layers 13–16 (L13H6, L13H8, L14H12, L14H13, L14H25, L15H5, L15H7, L15H14, L15H21, L15H22, L15H27, L16H25), although one was chosen by causal effect and the other without any patching.

Caveat. The six generic heads were identified from J-head readings and attention, not from patched accuracy. But the decision to remove them followed the FV split, which was patched on check-1 items of both halves. The no-generic sets are therefore not as cleanly held out as the others.

5.4 Very small sets for single tasks

Some tasks need very few heads (Table 8). These sets come from the exploratory runs, chosen and scored on the same items, so their numbers are optimistic. They show that a task can rest on one or a few heads, but no such head was found for translations, sequences, or present–past with rule-style means.

Task	Heads	How chosen	Check 1
country–capital	L16H25 alone	attends to *capital*, *country* in the instruction and decodes *city*	0.88
landmark–country	5	top 5 by attention to task words × decoding	0.79
product–company	20	same rule, top 20	0.72
singular–plural	20	same rule, top 20	0.68
present–past	L15H1 alone	found in the top-5 J-head set; example-style means only (check 2: 0.76)	0.80
landmark / product	5–10	Davidson et al.'s FV with fewer heads, α 3–4	0.95 / 0.87

Table 8: Small sets that carry one task (exploratory: chosen and scored on the same items, best α on them). Controls are low except product–company with the 5–10-head FV (0.22).

5.5 Check 2

In check 2 all sets lie close together (Figure 3, Table 5, App. A). Random sets already reach up to a quarter of the best sets' accuracy, and control means are larger than in check 1, on next- and prev-item as large as the task means. The ranking also differs: D, mid-table in check 1, is last here. With an example in the prompt the model needs little from the patch, so check 2 does not separate good sets from mediocre ones. We use check 1 to choose.

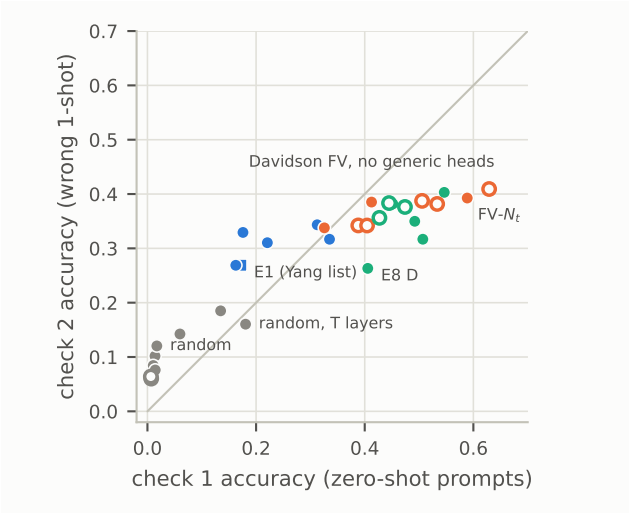


Figure 3: Check 2 against check 1, mean over the 14 tasks, every set of Table 5 run in both checks (colours and markers as Figure 1). The line is $y = x$. Check-1 accuracy spans 0 to 0.63, check-2 accuracy of the non-random sets only 0.26 to 0.41.

5.6 What did not help

- Heads that pass Yang et al.'s rule under all five templates: better only on fact tasks. Keeping only the union's heads shared across tasks removes most of its gains.
- Heads chosen on what the instruction adds (E6), and word lists without a contrast between tasks.
- Attention to the query (I) or to the rest of the instruction (R), in any combination.
- For shared sets, the weakest task's score instead of the average or the average rank.
- Head scores from GPT-5.5 reading each head's top tokens, run on two tasks only. On country–capital, the 20 J-head picks matched CC-20 and fell below FV-20. On English–German, nothing worked below 60 heads. J-head picks beat Logit-head picks on English–German at every size and on country–capital up to 30 heads.

6 Best approaches

For the stated goal, a small set of heads that represents the tasks in general:

1. **Read heads with J-head.** It finds more heads that carry the task, in middle layers the logit lens misses, and its lead holds at equal set size.
2. **One shared set: Davidson et al.'s FV without its generic heads** (25 heads). Rank heads by their causal score averaged over tasks, take the top 30, drop the heads whose J-head readings are alike on every task (line ends, colons), and add the summed means after layer 11 at $\alpha = 2$. It is the best set we found on average and on facts, and it carries every task type but sequences.
3. **One shared set patched head by head: FV-univ-30 or FV-univ-40**, or FV-univ-20 without its generic heads. Twenty heads with the generic ones included are not reliable.
4. **One shared set without patching: E10 shared DBT** at 30 to 40 heads. It needs only J-head readings and attention patterns, and it holds many of the same middle-layer heads as the causal sets.
5. **Per task, if size does not matter: FV-N_t** (about 120 heads), the only set that carries sequences and the best on morphology; or **T** (about 150), best of the per-task sets on facts and translations.
6. **Per task, small: DBT** (about 40 heads) or **DB** (about 65). Both are lens and attention only, and above causally ranked sets of similar size.
7. **By task type**: translations need attention-filtered heads or the FV without generic heads; sequences need many causally ranked heads per task; fact tasks can rest on one to five heads.

7 Limitations

One model and 14 simple tasks. Every run is single and greedy, without intervals, so differences of a few points between sets are not established. Sets differ in size, and size alone changes accuracy. Only α 1 and 2 are compared, which may underrate sets that peak higher: sequences peaked at larger α in earlier runs, and $\alpha = 2$ is chosen for about two thirds of the set and task pairs in check 1. The generic heads were removed after a result on the same items (Sec. 5.3). T uses Yang et al.'s term lists as part of its task words. The CC and E7 word lists predate the case fix of the decoding filter and miss capitalised name tokens. Items are held out, but wordings are not: every set was chosen and tested on the same five templates, so whether the sets carry the task when it is asked differently is untested. Check 2 has few items on several tasks.

Conditions still to run

E11 ran the conditions listed in the first version of this document: both folds for FV-univ, the shared rankings at 5 to 40 heads, random controls for every kind of set, and Davidson et al.'s intervention in both checks with and without its generic heads. Table 9 lists what remains, under the same protocol. None has been run.

Condition	What it settles
The 25 no-generic FV heads patched head by head	how much of the best set's lead comes from Davidson et al.'s intervention rather than the heads
The no-generic FV at fewer heads (top 10, 15, 20 of the ranking minus generic heads)	the smallest version of the best set
Generic heads identified on each choosing half	a cleanly held-out version of the no-generic sets
α up to 3–4 for the leading sets	whether sets that prefer $\alpha = 2$, and the sequences, gain further
Best rules under Logit-head (DB, DBT, E10 shared DBT)	the readout comparison under the common protocol, not only in E1 (word list for D to be decided)
Best sets on new wordings (E4b; wordings await a hand check)	whether the sets carry the task when it is asked differently

Table 9: Runs that would complete the comparison. All but the last reuse existing readings, scores and means.

Code and data. Tables and figures are built from the saved result files by `scripts/synthesis_report.py`; each experiment's own report gives its scripts.

References

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A Per-task results

Task	Items	Unp.	E1	CC	E7 +B	FV-20	FV-N_t	T	DB	DBT	FV-30	FV−gen	FV-univ	E10 DBT
Word relations														
antonym	100/100	0.00	0.00	0.42	0.46	0.02	0.49	0.60	**0.70**	0.61	0.25	0.62	0.26	0.08
synonym	100/100	0.03	0.10	0.34	0.30	0.18	**0.55**	0.18	0.38	0.27	0.04	0.33	0.22	0.15
Morphology														
singular–plural	91/88	0.00	0.00	0.69	0.88	0.44	0.85	0.80	0.78	0.80	0.69	**0.91**	0.77	0.73
present–past	100/100	0.00	0.00	0.04	0.17	0.64	**0.85**	0.00	0.54	0.00	0.05	0.53	0.69	0.15
Facts														
country–capital	84/86	0.00	0.58	0.75	0.87	0.88	0.94	0.98	0.69	0.94	0.77	**0.99**	0.95	0.94
country–currency	67/65	0.00	0.12	0.29	0.32	0.00	0.47	0.61	0.45	0.27	0.35	**0.70**	0.67	0.58
landmark–country	100/100	0.01	0.72	0.92	0.03	0.92	0.93	0.93	0.73	0.81	0.89	**0.95**	0.86	0.94
park–country	100/100	0.00	0.43	0.73	0.42	0.82	0.88	0.67	0.47	0.55	0.87	**0.89**	0.68	0.79
product–company	100/100	0.07	0.33	0.24	0.37	0.64	**0.91**	0.82	0.40	0.74	0.79	0.89	0.80	0.83
Sequences														
next-item	58/63	0.00	0.07	0.17	0.02	0.01	**0.70**	0.10	0.03	0.10	0.19	0.10	0.10	0.08
prev-item	53/50	0.00	0.04	0.13	0.16	0.01	**0.19**	0.04	0.04	0.03	0.04	0.15	0.14	0.08
Translations														
English–French	100/100	0.00	0.00	0.26	0.13	0.00	0.21	0.49	**0.57**	0.49	0.17	0.51	0.35	0.39
English–German	100/100	0.00	0.00	0.18	0.07	0.00	0.02	**0.67**	0.60	0.50	0.09	0.48	0.37	0.29
English–Spanish	100/100	0.00	0.00	0.38	0.17	0.01	0.25	0.76	0.71	**0.77**	0.24	0.75	0.59	0.58
Mean		0.01	0.17	0.40	0.31	0.33	0.59	0.55	0.51	0.49	0.39	0.63	0.53	0.47
Heads (median)			26	118	32	20	118	154	65	38	30	25	40	30

Table 10: Check 1 per task, held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. Items: patched items in fold 1 / fold 2. Unp.: unpatched accuracy. Columns as in Figure 2; E7 +B: E7 words + B; FV−gen: Davidson FV-30 without its generic heads; FV-univ: FV-univ-40; E10 DBT: E10 shared DBT (30). Bold: the row's highest value.

Task	Items	Unp.	E1	CC	E7 +B	FV-20	FV-N_t	T	DB	DBT	FV-30	FV−gen	FV-univ	E10 DBT
Word relations														
antonym	100/100	0.00	0.32	0.35	0.34	0.30	0.35	**0.37**	0.33	0.36	0.24	0.29	0.32	0.21
synonym	100/100	0.00	0.15	0.17	0.16	0.11	0.18	0.07	0.16	0.14	0.12	**0.22**	0.18	0.12
Morphology														
singular–plural	11/15	0.00	0.56	0.77	0.77	0.64	0.65	0.76	0.79	0.75	0.78	**0.83**	0.75	0.77
present–past	30/31	0.00	0.14	0.64	0.78	**0.89**	0.82	0.75	0.69	0.46	0.52	0.74	0.85	0.87
Facts														
country–capital	14/13	0.00	0.77	0.77	0.74	0.67	0.74	0.81	0.68	0.71	0.63	**0.82**	0.69	0.80
country–currency	31/34	0.00	0.12	0.23	0.12	0.19	0.17	0.29	0.22	0.16	0.25	**0.37**	0.21	0.23
landmark–country	100/87	0.00	0.34	0.32	0.19	0.33	0.32	0.31	0.22	0.34	**0.38**	0.35	0.33	0.32
park–country	22/23	0.00	0.18	0.33	0.38	0.33	0.38	0.38	0.22	0.26	0.29	0.35	**0.40**	0.33
product–company	84/75	0.00	0.19	0.16	0.21	0.20	0.22	**0.30**	0.16	0.22	0.26	0.25	0.24	0.21
Sequences														
next-item	54/50	0.00	0.34	0.34	0.25	0.23	**0.61**	0.44	0.12	0.55	0.22	0.43	0.34	0.38
prev-item	59/63	0.00	0.09	0.09	0.12	0.17	0.32	0.17	0.05	0.10	**0.34**	0.29	0.19	0.15
Translations														
English–French	100/100	0.00	0.22	0.16	0.22	0.26	**0.36**	0.35	0.25	0.29	0.33	0.30	0.29	0.29
English–German	100/100	0.00	0.19	0.21	0.27	0.21	0.17	**0.30**	0.27	0.24	0.20	0.23	0.29	0.28
English–Spanish	100/100	0.00	0.16	0.24	0.25	0.20	0.22	**0.35**	0.29	0.30	0.25	0.26	0.28	0.31
Mean		0.00	0.27	0.34	0.34	0.34	0.39	0.40	0.32	0.35	0.34	0.41	0.38	0.38
Heads (median)			26	118	32	20	118	154	65	38	30	25	40	30

Table 11: Check 2 per task, as Table 10. Few items on singular–plural, country–capital and park–country.